

Total No. of Questions : 8]

SEAT No. :

PE-2569

[Total No. of Pages : 3

**[6583]-99**  
**T.E. (E&TC)**  
**DIGITAL COMMUNICATION**  
**(2019 Pattern) (Semester - V) (304181)**

*Time : 2½ Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) *Answer any one Question out of Q. No. 1 or Q.2, Q. No. 3 or Q.4, Q.No.5 or Q.6 and Q. No.7 Or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Assume suitable data, if necessary.*

**Q1) a)** Explain M array QAM in following terms. **[8]**

- i) Draw signal space diagram of 16 - QAM
- ii) Comment on Euclidean distance for 16 QAM

**b)** Calculate bandwidth requirement and minimum separation (Euclidean distance) of signal points in signal space for

- i) 16-PSK
- ii) 16-FSK
- iii) 16-QAM.

Given that input bit rate is 8kbps and bit energy is  $1 \times 10^{-4}$  J. **[9]**

OR

**Q2) a)** Compare M-ary PSK and M-ary FSK. **[8]**

**b)** What is OFDM? Draw and Explain block diagram of generation and reception of the OFDM. **[9]**

**Q3) a)** Draw the block diagram of spread spectrum digital communication and explain the various blocks. **[9]**

**b)** A spread spectrum communication system has the following parameters. Information bit duration  $T_b = 4.095$  ms, PN chip duration ,  $T_c = 1$  microsecond. Find the processing gain. What is the number of shift registers required? Also find the jamming margin if  $E_b/N_0 = 10$  for BPSK Scheme. **[9]**

OR

**P.T.O.**

**Q4) a)** Draw and explain FHSS spread spectrum system with transmitter and receiver section. [9]

b) Design a three stage feedback shift register with proper taps to generate N=7 PN sequence. Draw the generator block and if the initial state of shift register is 100 (from left to right). Find the output sequence. [9]

**Q5) a)** Define all five Types of Entropies and Mutual Information in communication system. [8]

b) Apply Shannon-Fano code for following message ensemble and find coding efficiency [9]

$$X = [x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8]$$

$$P = [1/4, 1/8, 1/16, 1/16, 1/16, 1/4, 1/16, 1/8]$$

OR

**Q6) a)** An analog signal is band-limited to 4 kHz. It is sampled at the Nyquist rate and samples are quantized into 4 levels. The quantization levels  $Q_1, Q_2, Q_3$ , and  $Q_4$  are independent messages and have the probabilities  $P_1=P_2=1/8$  and  $P_3=P_4=3/8$ , Find the information rate of the source. [8]

b) A discrete memory less source has an alphabet of five symbols with their probabilities shown. Construct Huffman code and calculate code efficiency redundancy of the code. [9]

| Symbol      | $M_1$ | $M_2$ | $M_3$ | $M_4$ | $M_5$ |
|-------------|-------|-------|-------|-------|-------|
| Probability | 0.4   | 0.19  | 0.16  | 0.15  | 0.10  |

**Q7) a)** For a systematic linear block code, the three parity check digits are given by [9]

$$C_4 = d_1 \oplus d_2 \oplus d_3$$

$$C_5 = d_1 \oplus d_2$$

$$C_6 = d_1 \oplus d_3$$

i) Construct Generator Matrix

ii) Construct All Code generated by this matrix

iii) Determine error correcting capability

iv) Prepare suitable decoding table

v) Decode the received words 0 0 0 1 1 0

b) Define and Explain following terms;

[9]

- i) Hamming distance
- ii) Hamming weight
- iii) Code rate
- iv) Constraint length
- v) Generator polynomial

OR

**Q8)** a) A (7,4) cyclic code is described by a generator polynomial  $g(x) = 1 + x^2 + x^3$  [9]

- i) Find the codeword using polynomial division method for  $m = 1010$
- ii) Design an encoder for systematic code generation and explain its working

b) Explain Properties of Linear Block code and Cyclic Code with example [9]

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